Example 1: Pedestrian bridge over a highway

This example shows the process to identify the modal parameters of the pedestrian bridge over a highway located in municipality of Arucas (Gran Canaria, Canary Islands, Spain), at coordinates 28°09'07.0"N 15°31'33.1"W).

The vertical velocity measurements used on this example were taken at the points shown in Figure 1, and are grouped into 5 different sets formed by the points $S_1 = \{P_1, P_2\}$; $S_2 = \{P_2, P_3\}$; $S_3 = \{P_3, P_4\}$; $S_4 = \{P_4, P_5\}$; and $S_5 = \{P_5, P_6\}$.



Figure 1: Distribution of measurement points

The file "Run_example.m" included in this repository contains two main calls: one to the "download_zenodo_data.m" script, which downloads the necessary experimental data directly from the Zenodo repository [1], and another to the MatRROMA function:

```
matrroma("Lvl_crossing_RR.txt",50,512,1,"COVD",50,'downsampling_rate',64)
```

In this call, the arguments are the following:

- `filenames`: The file containing the names and the disposition of the sensors in the assembly is "Lvl_crossing_RR.txt".

```
Lvl_crossing_P1_P2.dat 1 1 0 90
                       2 0 90
Lvl_crossing_P2_P3.dat 2 2 0 90
                       3 0 90
Lvl_crossing_P3_P4.dat 3 3 0 90
                       4 0 90
Lvl_crossing_P4_P5.dat 4 4 0 90
                       5 0 90
Lvl_crossing_P5_P6.dat 5 5 0 90
                       6 0 90
```

Figure 2: Input file defining the disposition of the different sensor positions

  As observed in the global configuration file, each data file (with its own set ID) is associated with two sensor positions (channels). For each channel, the file lists its position ID number followed by its azimuth and elevation angles. In this case, the azimuth is 0° and the elevation is 90°, since all measurements were taken along the vertical axis. Note that each column in the data files corresponds to a channel, and each row represents a data sample.

- `q`: The selected number of block rows of the Hankel matrices is 50.
- `sampling_rate`: The sampling rate of the measurements was 512 Hz.
- `setref`: The set chosen as master set is S_1.

- `method`: The chosen identification method is "COVD".
- `max_order`: The maximum order displayed in the stabilization diagram is 50.
- `downsampling_rate`: Introduced as a parameter, meaning that the sampling rate used for the identification is 64 Hz, rather than the initial 512 Hz.

Executing the MatRROMA function opens the graphical user interface (GUI), shown in Figure 3, which allows the user to define the stabilization criteria and select the representative poles. In this example, the default stabilization criteria are maintained. The desired poles (2.91 Hz at order 36; 10.63 Hz at order 36; and 22.41 Hz at order 28) are chosen by left-clicking on them and pressing the "Save" button.

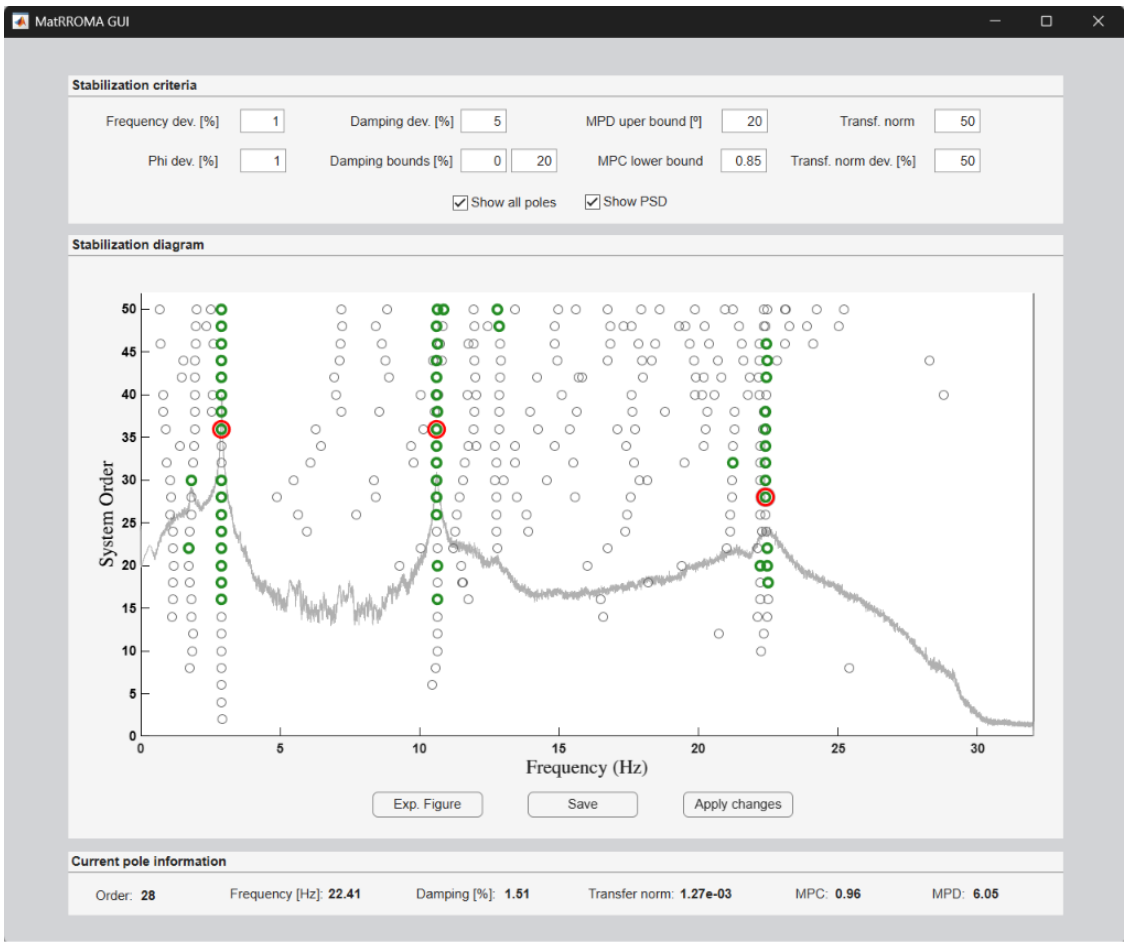


Figure 3: MatRROMA GUI

Saving this selection automatically generates a ".mat" file. For reference, the "Dynamic_Parameters.mat" file provided in this repository contains the final results of this illustrative case.

References

[1] S. González-Jiménez, L. A. Padrón, G. M. Álamo, and J. D. R. Bordón, "Dataset employed in operational modal analysis from multi-setup measurements with roving references and stochastic subspace identification," May 2025. DOI: https://doi.org/10.5281/zenodo.15545604.